FRE 6073 Derivatives – Midterm Study Guide

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The midterm is closed book on Oct 17, 2026 at 2pm in class and covers forwards, futures and swaps (Hull 11th ed., Ch 1–7). There is no class on Oct 10, and Assignment 2 is due Oct 10, 2026 at 11:59pm. Prof. Gupta's exam questions test whether you can apply the concepts to new problems, not repeat textbook ones, so practice setting up arbitrage arguments. Bring a calculator.

Week 1 – Introduction to derivatives

A derivative's value comes from an underlying asset. Know the four types, where they trade, and the three kinds of traders.

Material: Lecture 1 slides, Hull Ch 1.

- Four types: forwards, futures, swaps, options
- Exchange-traded (e.g. CME) vs OTC. Since 2008: standard OTC trades go through SEFs and are cleared by CCPs, and all trades are reported to a repository. Goals: less systemic risk, more transparency.
- Uses: hedge, speculate, lock in arbitrage, change the nature of a liability or investment
- Long = agreed to buy. Short = agreed to sell. Long forward payoff = S_T − K, short = K − S_T
- Forward price = the delivery price that makes the contract worth zero today
- Forward/futures = obligation. Option = right. Call = right to buy, put = right to sell. American = exercise any time, European = only at maturity.
- Forward price of a non-dividend stock (annual compounding): F = S(1 + r)^T. Example: S = 60, r = 5%, T = 1 → F = 63

Worked examples from the slides

Type	Setup	Answer
Hedge with forward	US firm owes £10M in 3 months	Go long a 3-month GBP forward to lock the rate
Hedge with puts	1,000 shares at $28, 2-month $27.50 put costs $1	Buy 10 contracts. Holding is floored at $27,500, cost $1,000

Type	Setup	Answer
Speculate	$2,000, stock $20, 2-month $22.50 call costs $1	100 shares or 2,000 calls. At $27: shares +$700, calls +$7,000. Below $22.50 the calls lose all $2,000
Arbitrage	£100 in London, $120 in NY, 1.2300 $/£	London price = $123. Buy in NY, sell in London: $3 per share

- Danger: hedgers can drift into speculating, so firms need controls (risk limits)
- Hedge funds avoid mutual-fund rules (disclosure, redemption, leverage limits). Strategies: long/short equity, convertible arb, distressed, emerging markets, global macro, merger arb.

Check yourself

- [ ] Draw long and short forward payoff diagrams
- [ ] Work the speculation example for a final price of $21

Week 2 – Forward and futures prices

Forward prices come from no-arbitrage cost of carry: buy the asset, finance it, and deliver it. Every formula is a version of $F_0 = S_0 e^{cT}$.

Material: Lecture 2 slides, Hull Ch 5.

Underlying	Forward price F_0	Notes
Investment asset, no income	$S_0 e^{rT}$	r = T-year continuous risk-free rate
Known income with PV I	$(S_0 - I) e^{rT}$	Hull Ch 5
Known yield q	$S_0 e^{(r-q)T}$	Stock index: q = dividend yield
Foreign currency	$S_0 e^{(r-r_f)T}$	Foreign rate acts like a yield
Consumption asset, storage u	$\leq S_0 e^{(r+u)T}$	Inequality because of convenience yield
General	$S_0 e^{(c-y)T}$	c = cost of carry, y = convenience yield

Value of an existing forward (delivery price K)

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f_{\text{long}} = (F_0 - K)e^{-rT}, \qquad f_{\text{long}} = S_0 - Ke^{-rT}
\;(\text{no income})
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- Practice Q1: S = 40, r = 5%, T = 1 → F0 = 40e^0.05 = 42.05, value 0. After 6 months, S = 45: new forward price 45e^0.025 = 46.14, contract value = 45 − 42.05e^(−0.025) = 3.99
- Practice Q2: forward = futures price when rates are deterministic (Hull's daily-resettlement proof). Positive correlation between rates and the asset makes futures slightly higher than forwards; negative makes them lower.

Index arbitrage

- F0 > S0 e^((r−q)T): buy the stocks, sell futures
- F0 < S0 e^((r−q)T): buy futures, short the stocks
- Needs the index to be a tradable portfolio. Usually done by computer; it can fail when simultaneous trades are impossible.

Investment vs consumption assets

- Investment assets (gold, silver) are held for investment, so the equality holds
- Consumption assets (copper, oil) are held for use. People won't short them to arbitrage, so only F0 ⩽ S0 e^((r+u)T) holds

Check yourself

- [ ] Rework Practice Q1 without notes
- [ ] Set up the cash-and-carry and reverse cash-and-carry trades
- [ ] Price a 6-month currency forward from two interest rates

Week 3 – Futures markets and clearing

Futures are standardized, exchange-traded and settled daily through margin accounts, which removes almost all credit risk.

Material: Lecture 3 slides, Hull Ch 2.

Margins

- Initial margin is posted at the start. If the balance falls below the maintenance margin, the trader must top it back up to the initial margin (the variation margin).
- Clearing house members keep the balance at the initial margin every day and also pay into a default fund

- Slide example: long 2 December gold contracts, 100 oz each, at $1,750. Initial margin $12,000 total, maintenance $9,000. On day 7 the balance hits $7,980, below $9,000, so the margin call is $4,020 (back to $12,000).
- Daily gain = (settle today − settle yesterday) × contract size × number of contracts

Terms

- Open interest = contracts outstanding (number of longs = number of shorts)
- Settlement price = price just before the close, used for daily settlement
- Volume = number of trades in a day
- Most contracts are closed out with an offsetting trade before maturity. If delivered, the short chooses what, where and when. Index futures are cash-settled.

Forwards	Futures
Private, OTC	Exchange-traded
Non-standard	Standard
Usually one delivery date	Range of delivery dates
Settled at the end	Settled daily
Delivery or final cash settlement usually happens	Usually closed out before maturity
Some credit risk	Virtually no credit risk

OTC clearing

- Bilateral trades: ISDA Master Agreement plus a credit support annex (CSA) that sets collateral. Traditionally variation margin only, no initial margin (LTCM example).
- After the crisis: standard trades between financial institutions must be cleared through CCPs. Non-standard trades between them must post both initial margin (with a third party) and variation margin. Trades with non-financial corporates can stay bilateral.

Check yourself

- [ ] Rebuild the gold margin table and find when each margin call happens
- [ ] List four differences between forwards and futures

Week 4 – Interest rates

A rate means nothing without its compounding frequency. Zero rates are bootstrapped from bond prices one maturity at a time.

Material: Lecture 4 slides, Hull Ch 4.

Reference rates

- Treasury rates are seen as artificially low (no bank capital needed, tax advantages). SOFR and similar rates are now the risk-free proxies.
- Fed funds = unsecured overnight interbank rate. The effective rate is a volume-weighted median.
- Repo = sell a security and agree to buy it back: a collateralized loan. The rate comes from the price difference.
- LIBOR (unsecured AA bank borrowing estimate) was phased out by end-2021 after manipulation concerns. Replacements are compounded overnight rates: SOFR (USD), SONIA (GBP), €STR (EUR), SARON (CHF), TONAR (JPY).
- SOFR = volume-weighted median of three Treasury repo segments: BNY Mellon tri-party GC, FICC GCF, and FICC DVP bilateral (lowest 25% of volume trimmed to drop "specials"). Median is more robust than a mean and is a rate actually traded.

Compounding

- $100 at 10% for a year: annual 110.00, semiannual 110.25, quarterly 110.38, monthly 110.47, daily 110.52
- Continuous: A grows to Ae^(RT), discounts by e^(−RT)

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R_c = m\ln\left(1 + \frac{R_m}{m}\right), \qquad R_m = m\left(e^{R_c/m} - 1\right)
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- Examples: 10% semiannual = 2 ln(1.05) = 9.758% continuous. 8% continuous = 4(e^0.02 − 1) = 8.08% quarterly.

Zero rates and bootstrapping

- Zero (spot) rate for T = return on an investment that pays only at T
- Slide data (principal 100): 0.25y price 99.6, 0.5y 99.0, 1y 97.8 (zeros), 1.5y 4% coupon at 102.5, 2y 5% coupon at 105.0
- Short end directly: R = ln(100/price)/T → 1.603%, 2.010%, 2.225%

- 1.5 years: solve 2e^(−0.02010×0.5) + 2e^(−0.02225×1) + 102e^(−R×1.5) = 102.5 → R = 2.284%. The 2-year rate is 2.416% the same way.

Check yourself

- [ ] Convert 6% monthly compounding to continuous and to annual
- [ ] Bootstrap the 2-year zero rate from the slide data
- [ ] Explain why SOFR uses a volume-weighted median

Week 5 – Bonds, duration and swaps

A swap is a bundle of forward contracts. Value it by assuming forward rates are realized, then discounting the net cash flows at the risk-free (SOFR) rates. Several formulas on these slides were images that did not come through, so the standard Hull forms are given here.

Material: Lecture 5 slides, Hull Ch 4 and Ch 7.

Forward rates and bonds

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R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}, \qquad R_F^{\text{inst}} = R + T\frac{\partial R}{\partial T}
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- Bond price = each cash flow discounted at its own zero rate. Slide example: 2-year 6% semiannual coupon → $98.39. That figure uses Hull's older zero rates (5.0%, 5.8%, 6.4%, 6.8%). With the Week 4 zero curve the same bond is worth about $106.94, so ask which one the class used
- Bond yield y = the single rate that gives the market price. Example: 6.76% continuous
- Par yield = coupon rate that makes price = face value: c = (100 − 100d) m / A, where d = discount factor at maturity and A = sum of coupon-date discount factors

Duration and convexity

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D = \frac{\sum t_i c_i e^{-yt_i}}{B}, \qquad \frac{\Delta B}{B} \approx -D\,\Delta y + \tfrac{1}{2}C(\Delta y)^2, \qquad C = \frac{\sum c_i t_i^2 e^{-yt_i}}{B}
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- With yield compounded m times a year, use modified duration D* = D / (1 + y/m)
- Duration hedging protects only against small parallel shifts. Matching asset and liability durations still leaves non-parallel (twist) risk. Convexity covers larger shifts, still parallel.

Interest rate swaps

- Plain vanilla example: Apple receives 3-month SOFR and pays 3% on $100M for 2 years with Citi
- Uses: turn a floating liability into fixed (Apple pays SOFR + 0.1% on a loan → net 3.1% fixed), fixed liability into floating (Intel 3.2% → SOFR + 0.23%), and the same for assets
- Day counts: SOFR usually actual/360. Fixed leg actual/365 or 30/360
- ISDA Master Agreements govern terms. CCPs clear standard swaps between financial institutions.

Valuing a swap (slide example)

Pay 3%, receive SOFR, semiannual, $100M, 1.2 years left (0.2, 0.7, 1.2). Zero rates 2.8%, 3.2%, 3.4%. Last 0.3 years of SOFR = 2.3%.

Time (yrs)	Fixed ($M)	Floating ($M)	Net ($M)	Discount factor	PV ($M)
0.2	−1.500	+1.258	−0.242	0.9944	−0.241
0.7	−1.500	+1.694	+0.194	0.9778	+0.190
1.2	−1.500	+1.857	+0.357	0.9600	+0.343
Total					+0.292

- First floating rate: 0.6 × 2.3% + 0.4 × 2.8% = 2.50% continuous = 2.516% semiannual. Forward rates 3.36% (0.2–0.7) and 3.68% (0.7–1.2), continuous

Currency swaps

- Fixed-for-fixed: exchange principals at start and end, pay interest in each currency. BP example: pay 3% on $15M, receive 4% on £10M for 5 years
- Value two ways: as the difference of two bonds, or as a series of FX forwards
- Slide example: receive 3% yen, pay 4% USD, $10M and ¥1,200M, 3 years, JPY 1.5%, USD 2.5%, 110 ¥/$. Yen bond = ¥1,252.01M, USD bond = $10.4191M → value = 1,252.01/110 − 10.4191 = +$0.9629M (both methods agree)
- Fixed-for-floating = fixed-for-fixed currency swap + interest rate swap

Credit default swaps (first look)

- Protection buyer pays the CDS spread (e.g. 150 bp) on the notional. On default the seller buys the reference bonds at face value and the spread payments stop.
- Other swaps: amortizing, step-up, compounding, quanto, equity, extendible/puttable, commodity, volatility

Check yourself

- [ ] Recompute the 0.7-year forward rate and its floating cash flow
- [ ] Value the yen/dollar swap with forward FX rates
- [ ] Estimate a bond's price change for a 50 bp yield rise using D and C

Also on the reading list: Hull Ch 3 and Ch 6

The midterm announcement lists "Hedging strategies using futures" and "Interest rate futures", but no slides for them are posted yet. Check whether they were covered in class. The standard Hull results are below (from the textbook, not the course slides).

Ch 3 – Hedging with futures

- Basis = spot price − futures price. Basis risk remains when the hedge asset or date doesn't match exactly.
- Short hedge if you will sell the asset, long hedge if you will buy it
- Minimum variance hedge ratio $h^* = \rho \, \sigma_S / \sigma_F$. Number of contracts $N^* = h^* Q_A / Q_F$
- Stock index hedge: $N^* = \beta \, V_A / V_F$. To move beta from β to β^*: $N = (\beta^* - \beta) \, V_A / V_F$ (negative = short)
- Rolling a hedge forward: close near contracts and open later ones (Metallgesellschaft is the warning case)

Ch 6 – Interest rate futures

- Day counts: actual/actual (Treasury bonds), 30/360 (corporates), actual/360 (money market)
- Cash (dirty) price = quoted (clean) price + accrued interest
- Treasury bond futures: the short delivers the cheapest-to-deliver bond. Cash received = quoted futures price × conversion factor + accrued interest
- Duration-based hedge ratio: $N^* = P \, D_P / (V_F \, D_F)$